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Friction weld plug

Abstract

A friction weld plug includes a gasket and a stud. The stud includes a nose configured to be inserted into a first hole formed within a wall. The nose includes a cavity having female threads. The stud also includes a shoulder extending radially from the nose and including a groove configured to contain the gasket. The friction weld plug also includes a spacer including a second hole and a fastener including male threads. The fastener is configured to be inserted through the second hole, into the first hole, and into the cavity such that the male threads and the female threads form an attachment that presses the spacer against a first side of the wall and presses the gasket against a second side of the wall that is opposite the first side, thereby forming a seal between the gasket and the second side of the wall.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11577338	12/2022	Oezkan	N/A	B23K 20/1295
2015/0222027	12/2014	Dupuis	439/805	H01R 4/029

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Background/Summary

FIELD

(1) The present disclosure generally relates to a friction weld plug, and more specifically to a friction weld plug that includes a gasket, a stud, a spacer, and a fastener.

BACKGROUND

(2) Using friction stir welding to weld two sections of material typically yields a hole on the weld line where the friction weld tool is removed from the material at the end of the weld line. The hole is generally filled with a plug which usually causes degraded material properties at the vicinity of the hole when compared to the rest of the weld line. Thus, the material is often designed such that the degraded material properties at the hole meets a manufacturing standard and the rest of the weld line greatly exceeds that standard, which can lead to some inefficiencies. As such, a need exists for a friction weld plug that provides better material properties for the weld, such as increased strength and ductility.

SUMMARY

(3) One aspect of the disclosure is a friction weld plug comprising: a gasket; a stud comprising: a nose configured to be inserted into a first hole formed within a wall, wherein the nose comprises a cavity having female threads; and a shoulder extending radially from the nose and comprising a groove configured to contain the gasket; a spacer comprising a second hole; and a fastener comprising male threads, wherein the fastener is configured to be inserted through the second hole, into the first hole, and into the cavity such that the male threads and the female threads form an attachment that presses the spacer against a first side of the wall and presses the gasket against a second side of the wall that is opposite the first side, thereby forming a seal between the gasket and the second side of the wall.

(4) Another aspect of the disclosure is a method for installing a friction weld plug, the method comprising: bonding, via a friction stir welding process, a first panel to a second panel, thereby forming a wall and producing a hole in the wall; inserting, from a first side of the wall, a fastener of

the friction weld plug through a spacer of the friction weld plug and into the hole in the wall; inserting, from a second side of the wall that is opposite the first side, a nose of a stud of the friction weld plug into the hole; and creating, via rotating male threads of the fastener against female threads of a cavity within the nose, an attachment that presses the spacer against the first side of the wall and presses a gasket against the second side of the wall, thereby forming a seal between the gasket contained in a groove in a shoulder of the stud and the second side of the wall.

(5) Another aspect of the disclosure is a friction weld plug comprising: a first spring seal; a second spring seal; a stud comprising: a nose configured to be inserted into a first hole formed within a wall, wherein the nose comprises a cavity having female threads; and a shoulder extending radially from the nose and comprising a first groove configured to contain the first spring seal and a second groove configured to contain the second spring seal; a spacer comprising a second hole; and a fastener comprising male threads, wherein the fastener is configured to be inserted through the second hole, into the first hole, and into the cavity such that the male threads and the female threads form an attachment that presses the spacer against a first side of the wall and presses the first spring seal and the second spring seal against a second side of the wall that is opposite the first side, thereby forming a first seal between the first spring seal and the second side of the wall and a second seal between the second spring seal and the second side of the wall.

(6) By the term “about” or “substantially” with reference to amounts or measurement values described herein, it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

(7) The features, functions, and advantages that have been discussed can be achieved independently in various examples or may be combined in yet other examples further details of which can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The novel features believed characteristic of the illustrative examples are set forth in the appended claims. The illustrative examples, however, as well as a preferred mode of use, further objectives and descriptions thereof, will best be understood by reference to the following detailed description of an illustrative example of the present disclosure when read in conjunction with the accompanying Figures.
- (2) FIG. 1 is a block diagram of a friction weld plug, according to an example.
- (3) FIG. 2 is a perspective view of a stud, according to an example.
- (4) FIG. 3 is a cross sectional view of a stud, according to an example.
- (5) FIG. 4 is a cross sectional view of a wall and a friction weld plug, according to an example.
- (6) FIG. 5 is a cross sectional view of a wall and a friction weld plug, according to an example.
- (7) FIG. 6 is a cross sectional view of a friction weld plug, according to an example.
- (8) FIG. 7 is a front view of a wall and a friction weld plug, according to an example.
- (9) FIG. 8 is a front view of a wall and a friction weld plug, according to an example.
- (10) FIG. 9 is a perspective view of a friction stir welding process, according to an example.
- (11) FIG. 10 is a block diagram of a method, according to an example.
- (12) FIG. 11 is a block diagram of a method, according to an example.

DETAILED DESCRIPTION

(13) As noted above, a need exists for a friction weld plug that provides better material properties for the weld. As such, an aspect of the disclosure is a friction weld plug that includes a gasket (e.g., a spring seal) and a stud. The stud includes a nose configured to be inserted into a first hole formed

within a wall. The first hole can be created by a friction stir welding process upon removal of the bit from the material. For example, the wall can be part of a tank used for storing pressurized gas and can take the form of two metallic panels welded together by the friction stir welding process. The nose includes a cavity having female threads. The stud also includes a shoulder extending radially from the nose and including a groove configured to contain the gasket. The friction weld plug also includes a spacer including a second hole and a fastener that includes male threads. The fastener is configured to be inserted through the second hole, into the first hole, and into the cavity such that the male threads and the female threads form an attachment that presses the spacer against a first side (e.g., an exterior side) of the wall and presses the gasket against a second side (e.g., an interior side) of the wall that is opposite the first side, thereby forming a seal between the gasket and the second side of the wall.

(14) Disclosed examples will now be described more fully hereinafter with reference to the accompanying Drawings, in which some, but not all of the disclosed examples are shown. Indeed, several different examples may be described and should not be construed as limited to the examples set forth herein. Rather, these examples are described so that this disclosure will be thorough and complete and will fully convey the scope of the disclosure to those skilled in the art.

(15) FIG. 1 is a block diagram of a friction weld plug **100**. The friction weld plug **100** includes a gasket **102A**, a gasket **102B**, and a stud **104**. The stud **104** includes a nose **106** and a shoulder **116**. The friction weld plug **100** also includes a spacer **120**, a fastener **124**, and a spring washer **158**.

(16) FIG. 2 is a perspective view of the stud **104**. As shown, the stud **104** includes the nose **106** that further includes a cavity **112** having female threads **114**. The cavity **112** is elongated in an axial direction and is substantially cylindrical. The nose **106** extends in an axial direction away from the shoulder **116** and includes a surface **142** and a surface **143** that face radially outward and are round. The nose **106** also includes a surface **144** and a surface **145** that face radially outward and are substantially flat. The nose **106** also includes a surface **152** that faces axially away from the shoulder **116** and that surrounds an opening to the cavity **112**.

(17) The stud **104** includes the shoulder **116** that extends radially from the nose **106** and includes a groove **118A** configured to contain the gasket **102A** and a groove **118B** that is configured to contain the gasket **102B**. The groove **118B** surrounds the groove **118A**. Both the groove **118A** and the groove **118B** have an annular shape, but other examples are possible. The shoulder **116** includes a surface **160** that faces radially outward and is round. The shoulder **116** also includes an outlet **140** within a surface **134** that is discussed in more detail below.

(18) The stud **104** is typically a singular machined metal part. That is, the nose **106** and the shoulder **116** are generally integrated.

(19) FIG. 3 is a cross sectional view of the stud **104**. The shoulder **116** includes the surface **134** that includes the groove **118A** and the groove **118B**. The shoulder **116** also includes a surface **136** opposite the surface **134** and a port **138** that provides fluid communication between the surface **134** and the surface **136**. The port **138** includes the outlet **140** between the groove **118A** and the groove **118B** and on the surface **136**. The port **138** can be used to test a seal created by the friction weld plug **100**. The gasket **102A** and the gasket **102B** can each take the form of a spring seal coated with a friction reducing material.

(20) FIG. 4 is a cross sectional view of the friction weld plug **100** and a wall **110**. As shown, the nose **106** is configured to be inserted into a hole **108** formed within the wall **110**. The friction weld plug **100** also includes the spacer **120** that includes a hole **122** and a fastener **124** that includes male threads **126**. The fastener **124** is configured to be inserted through the hole **122**, into the hole **108**, and into the cavity **112** such that the male threads **126** and the female threads **114** form an attachment that presses the spacer **120** against a side **128** of the wall **110** and presses the gasket **102A** and the gasket **102B** against a side **130** of the wall **110** that is opposite the side **128**, thereby forming a seal **132A** between the gasket **102A** and the side **130** of the wall **110** and forming a seal **132B** between the gasket **102B** and the side **130** of the wall **110**.

(21) As shown, the nose **106** is inserted into the hole **108** such that the surface **134** of the shoulder **116** contacts the side **130** of the wall **110**.

(22) The wall **110** can be part of a metallic tank used for storing pressurized gas and can take the form of two or more metallic panels welded together by a friction stir welding process. More specifically, the hole **108** can be created by the removal of the welding tool from the wall **110**.

(23) The spacer **120** generally takes the form of a metal plate having the hole **122** centered within the spacer **120**. The fastener **124** is typically a threaded bolt.

(24) FIG. 5 is a cross sectional view of the wall **110** and the friction weld plug **100**. The embodiment of the friction weld plug **100** shown in FIG. 5 includes a spacer **120** that is different from than the spacer **120** shown in FIG. 4. In FIG. 5, the spacer **120** includes a receptacle **146** that is configured to receive the nose **106** of the stud **104**. Otherwise, any component shown in FIG. 5 generally can have the same properties or structures as described above with reference to FIG. 4 above.

(25) As shown, the surface **152** of the nose contacts a surface **156** of the receptacle. The surface **156** faces axially and is configured to contact the surface **152** while the attachment exists between the male threads **126** and the female threads **114**.

(26) FIG. 6 is a cross sectional view of the friction weld plug **100** shown in FIG. 5. The spacer **120** includes the receptacle **146** that forms a surface **148**, a surface **149**, a surface **150**, and a surface **151**. The surface **148** and the surface **149** face radially inward and are round. The surface **150** and the surface **151** face radially inward and are substantially flat.

(27) The nose **106** is configured to be inserted into the receptacle **146** such that the surface **145** contacts the surface **150** and the surface **144** contacts the surface **151**, thereby causing the stud **104** to be rotationally fixed to the spacer **120**.

(28) FIG. 7 is a front view of the wall **110** and the friction weld plug **100** from above the wall **110**. As shown, the friction weld plug **100** can include one or more spring washers **158** between a head of the fastener **124** and the spacer **120**. The fastener **124** is configured to be inserted through the one or more spring washers **158** and the hole **122**, into the hole **108**, and into the cavity **112** such that the male threads **126** and the female threads **114** form the attachment. In FIG. 7, the stud **104** is obscured by the wall **110**.

(29) FIG. 8 is a front view of the wall **110** and the friction weld plug **100** from below the wall **110**. In FIG. 8, the spacer **120** and the fastener **124** are obscured by the wall **110**.

(30) FIG. 9 is a perspective view of a friction stir welding process. For example, a friction welding tool **402** is rotated at high speed and then lowered to make contact with a first panel **302** and a second panel **304** along a weld line **305**. More specifically, a pin **403** of the friction welding tool **402** penetrates into the first panel **302** and the second panel **304** and spins, whereas a shoulder **404** of the friction welding tool **402** generally spins on the surface of the first panel **302** and the second panel **304** to create a bond at the weld line **305**. The friction stir welding process causes the first panel **302** to bond with the second panel **304**, thereby forming the wall **110** and producing the hole **108** in the wall **110** when the friction welding tool **402** is removed from the wall **110**.

(31) Referring to FIG. 4 or FIG. 5, the fastener **124** is inserted through the spacer **120** and into the hole **108** from the side **128** of the wall **110**. Additionally, the nose **106** of the stud **104** is inserted into the hole **108** from the side **130** of the wall **110**. Also, the attachment that presses the spacer **120** against the side **128** of the wall **110** and presses the gasket **102A** and the gasket **102B** against the side **130** of the wall **110** is created via rotating the male threads **126** of the fastener **124** against the female threads **114** of the cavity **112**, thereby forming the seal **132A** between the gasket **102A** contained in the groove **118A** and the side **130** of the wall **110** and forming the seal **132B** between the gasket **102B** contained in the groove **118B** and the side **130** of the wall **110**.

(32) In some examples, the hole **108** is enlarged via drilling such that a diameter of the hole **108** matches (e.g., is substantially equal to) the diameter of the nose **106** of the stud **104**.

(33) FIG. 10 and FIG. 11 are block diagrams of a method **200** and a method **250** for installing the

friction weld plug **100**. As shown in FIG. **10** and FIG. **11**, the method **200** and the method **250** include one or more operations, functions, or actions as illustrated by blocks **202**, **204**, **206**, **208**, and **210**. Although the blocks are illustrated in a sequential order, these blocks may also be performed in parallel, and/or in a different order than those described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon the desired implementation.

(34) At block **202**, the method **200** includes bonding, via a friction stir welding process, the first panel **302** to the second panel **304**, thereby forming the wall **110** and producing the hole **108** in the wall **110**. This functionality is described above with reference to FIG. **9**.

(35) At block **204**, the method **200** includes inserting, from the side **128** of the wall **110**, the fastener **124** of the friction weld plug **100** through the spacer **120** of the friction weld plug **100** and into the hole **108** in the wall **110**. This functionality is described above with reference to FIG. **4** and FIG. **5**.

(36) At block **206**, the method **200** includes inserting, from the side **130** of the wall **110** that is opposite the side **128**, the nose **106** of the stud **104** of the friction weld plug **100** into the hole **108**. This functionality is described above with reference to FIG. **4** and FIG. **5**.

(37) At block **208**, the method **200** includes creating, via rotating the male threads **126** of the fastener **124** against the female threads **114** of the cavity **112** within the nose **106**, the attachment that presses the spacer **120** against the side **128** of the wall **110** and presses the gasket **102A** against the side **130** of the wall **110**, thereby forming the seal **132A** between the gasket **102A** contained in the groove **118A** in the shoulder **116** of the stud **104** and the side **130** of the wall **110**. This functionality is described above with reference to FIG. **4** and FIG. **5**.

(38) At block **210**, the method **250** includes enlarging the hole **108** in the wall via drilling. This functionality is described above with reference to FIG. **4** and FIG. **5**.

(39) The description of the different advantageous arrangements has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the examples in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different advantageous examples may describe different advantages as compared to other advantageous examples. The example or examples selected are chosen and described in order to explain the principles of the examples, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various examples with various modifications as are suited to the particular use contemplated.

Claims

1. A friction weld plug comprising: a gasket; a stud comprising: a nose configured to be inserted into a first hole formed within a wall, wherein the nose comprises a cavity having female threads; and a shoulder extending radially from the nose and comprising a groove configured to contain the gasket; a spacer comprising a second hole; and a fastener comprising male threads, wherein the fastener is configured to be inserted through the second hole, into the first hole, and into the cavity such that the male threads and the female threads form an attachment that presses the spacer against a first side of the wall and presses the gasket against a second side of the wall that is opposite the first side, thereby forming a seal between the gasket and the second side of the wall.
2. The friction weld plug of claim 1, wherein the groove is a first groove, the friction weld plug further comprising: a second gasket, wherein the shoulder further comprises a second groove configured to contain the second gasket, wherein the fastener is configured to be inserted through the second hole, into the first hole, and into the cavity such that the male threads and the female threads form the attachment that presses the second gasket against the second side of the wall, thereby forming a second seal between the second gasket and the second side of the wall.
3. The friction weld plug of claim 2, wherein the second groove surrounds the first groove.

4. The friction weld plug of claim 2, wherein the shoulder comprises: a first surface that includes the first groove and the second groove; a second surface opposite the first surface; and a port that provides fluid communication between the first surface and the second surface.
5. The friction weld plug of claim 4, wherein the port includes an outlet between the first groove and the second groove.
6. The friction weld plug of claim 1, wherein the gasket comprises a spring seal.
7. The friction weld plug of claim 1, wherein the gasket comprises a coating configured for friction reduction.
8. The friction weld plug of claim 1, wherein the nose comprises a surface that faces radially outward and is round.
9. The friction weld plug of claim 8, wherein the surface is a first surface and the nose comprises a second surface that faces radially outward and is substantially flat.
10. The friction weld plug of claim 9, wherein the spacer comprises a receptacle forming a third surface that faces radially inward and is round and forming a fourth surface that faces radially inward and is substantially flat.
11. The friction weld plug of claim 10, wherein the nose is configured to be inserted into the receptacle such that the first surface contacts the third surface and the second surface contacts the fourth surface, thereby causing the stud to be rotationally fixed to the spacer.
12. The friction weld plug of claim 10, wherein the nose forms a fifth surface that faces axially and that surrounds an opening to the cavity.
13. The friction weld plug of claim 12, wherein the receptacle forms a sixth surface that faces axially and is configured to contact the fifth surface while the attachment exists between the male threads and the female threads.
14. The friction weld plug of claim 1, further comprising a spring washer, wherein the fastener is configured to be inserted through the spring washer and the second hole, into the first hole, and into the cavity such that the male threads and the female threads form the attachment.
15. The friction weld plug of claim 1, wherein the nose extends in an axial direction away from the shoulder.
16. The friction weld plug of claim 1, wherein the shoulder comprises a surface that faces radially outward and is round.
17. A method for installing a friction weld plug, the method comprising: bonding, via a friction stir welding process, a first panel to a second panel, thereby forming a wall and producing a hole in the wall; inserting, from a first side of the wall, a fastener of the friction weld plug through a spacer of the friction weld plug and into the hole in the wall; inserting, from a second side of the wall that is opposite the first side, a nose of a stud of the friction weld plug into the hole; and creating, via rotating male threads of the fastener against female threads of a cavity within the nose, an attachment that presses the spacer against the first side of the wall and presses a gasket against the second side of the wall, thereby forming a seal between the gasket contained in a groove in a shoulder of the stud and the second side of the wall.
18. The method of claim 17, further comprising enlarging the hole in the wall via drilling.
19. The method of claim 18, wherein enlarging the hole in the wall via drilling comprises enlarging the hole such that the hole has a first diameter that is substantially equal to a second diameter of the nose.
20. A friction weld plug comprising: a first spring seal; a second spring seal; a stud comprising: a nose configured to be inserted into a first hole formed within a wall, wherein the nose comprises a cavity having female threads; and a shoulder extending radially from the nose and comprising a first groove configured to contain the first spring seal and a second groove configured to contain the second spring seal; a spacer comprising a second hole; and a fastener comprising male threads, wherein the fastener is configured to be inserted through the second hole, into the first hole, and into the cavity such that the male threads and the female threads form an attachment that presses

the spacer against a first side of the wall and presses the first spring seal and the second spring seal against a second side of the wall that is opposite the first side, thereby forming a first seal between the first spring seal and the second side of the wall and a second seal between the second spring seal and the second side of the wall.
